

Piyush Paul

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Summary

Motivated Computer Science undergraduate experienced in AI-powered SaaS platforms, cybersecurity tools, and scalable backend systems. Skilled in full-stack development, cloud deployment, automation, RESTful APIs, and AI integrations with strong interest in cybersecurity and autonomous AI agents.

Experience

Backend Developer Intern, Fluencer Market (Remote) Dec 2024 – Apr 2025

- Developed Flask REST APIs supporting influencer-brand workflows.
- Reduced backend latency by 35% through optimized queries and data modeling.
- Delivered scalable backend services with cross-functional teams.

Projects

PacketForge — DPDK Network Scrubber May 2026
C, DPDK, Python, FastAPI, Next.js, QEMU/KVM, Lua

- Built a high-performance DDoS mitigation appliance using DPDK, scrubbing 200K–470K pps with zero kernel overhead.
- Implemented dual-NIC scrubber pipeline with real-time telemetry dashboard, hot-reload Lua rules, and CLI-driven CIDR blocking.

ARTA — Autonomous Red Team Agent Apr 2026
FastAPI, Next.js, Gemini AI, SQLite, Puppeteer, Nmap, Metasploit

- Built AI-driven autonomous penetration testing pipeline with CVE analysis and exploit validation.
- Developed live dashboard using SSE for real-time tracking and findings visualization.

AstroRoni Sep 2025 – Apr 2026
Next.js, Flask, MongoDB, Razorpay, OpenAI, GCP

- Developed AI-powered astrology SaaS generating Kundli and numerology reports.
- Integrated secure authentication, payments, and scalable cloud microservices.

Technical Skills

Languages: JavaScript, TypeScript, Python, C, C++, Java, Rust, SQL

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Flask, FastAPI, Supabase, MongoDB

Cloud/DevOps: GCP, Docker, Git, QEMU/KVM

AI/ML: LLM Integration (OpenAI, Gemini, Anthropic), Prompt Engineering, Agentic AI, Multi-Agent Systems, RAG, AI API Orchestration, LangChain

Achievements

Synchronicity S2 — 1st Runner-Up 2026

- Built **PacketForge**, a DPDK-based DDoS scrubber appliance capable of processing 470K pps at line rate.

MLH Hackathon — 1st Runner-Up 2026

- Built **NeighbourNet**, a mesh networking solution for offline disaster communication.

MLH Hackdays — Winner 2026

- Won with **ARTA**, an AI-powered autonomous penetration testing system.

Frontend Hackathon — 2nd Place 2024

- Secured 2nd place among thousands of participants for frontend innovation.

Education

JIS College of Engineering, West Bengal, India Jun 2025 – Present
Bachelor of Technology in Computer Science — SGPA: 7.9/10
Coursework: DSA, Operating Systems, OOP, DBMS